

Mehmet Ergezer
Wentworth Institute of Technology
Boston, MA, USA
ergezerm@wit.edu

June 10, 2026

Editors
Experimental Mathematics

Dear Editors,

I am pleased to submit the manuscript “Witness-split + window-cardinality refinement for $r_3(N)$: architecture, empirical results, and a structural hard pocket” for consideration in *Experimental Mathematics*.

The manuscript reports a reproducible computational campaign around the frontier case $r_3(212)$, combining witness-based splitting, OEIS A003002 window-cardinality pruning, recursive refinement, CP-SAT, HiGHS MIP, CDCL/SAT, independently verified DRAT certificates, and a Lean formal-proof-search encoding for the remaining two hard benchmark instances. The paper does not claim to prove $r_3(212) = 43$; instead it provides a reproducible architecture, a tiered benchmark release, and a precise empirical characterization of the residual hard pocket. The associated dataset is available at [10.5281/zenodo.20463334](https://zenodo.org/record/20463334), and the preprint is available as [arXiv:2606.04016](https://arxiv.org/abs/2606.04016).

The work is not under consideration at another journal. The author takes full responsibility for the manuscript, code, computations, and all claims made in the submission.

Disclosure of generative-AI use. In accordance with the Taylor & Francis AI Policy, I disclose that in the preparation of this work I used Anthropic Claude (Claude Sonnet 4.5 and Claude Opus 4.6, accessed via the Claude Code agent interface) and OpenAI Codex (GPT-5, accessed via the Codex agent interface) for drafting and revision assistance on the manuscript, repository scaffolding, and computational orchestration. I reviewed and verified every claim and every line of code before submission, and I take full responsibility for the content. These tools did not generate mathematical claims, did not direct the campaign’s experimental design, and do not appear as authors.

Thank you for considering this submission.

Sincerely,

Mehmet Ergezer